Prototypes, Visualisations and Evaluation:

1. Dashboard Prototype:

The final prototype was developed as an analytical dashboard to support the interpretation of Amazon app reviews. The purpose of the dashboard was to combine anomaly detection and sentiment analysis into one clear output so that users could understand both the reliability and emotional tone of those reviews. This directly supports the project aim of improving insights quality in Amazon app reviews.

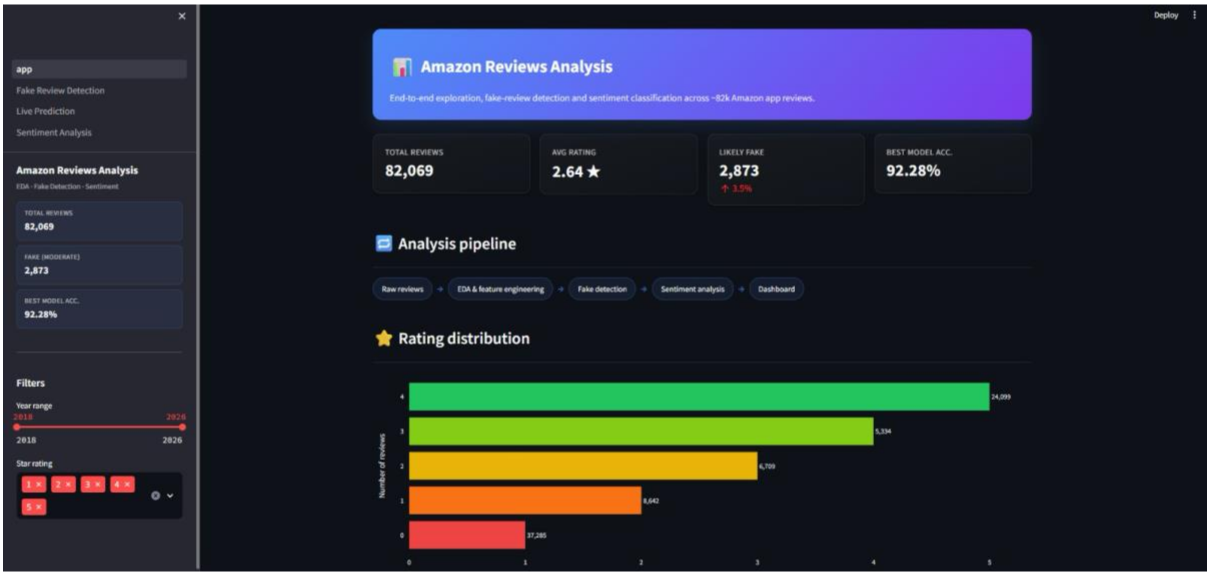

Figure 1: 7.1 Dashboard display

The dashboards show the main output including rating distribution, suspicious review detection, anomaly patterns sentiment classification results, and model evaluation results. The prototype allows the review data to be explore visually rather than only though numerical outputs. This makes the system more useful for decision making because suspicious patterns, review sentiment, and model performance can be understood quickly.

The dashboard demonstrates how the model could be used in practically by allowing the review text to be processed through the fake review detection logic and trained sentiment model.

2. Exploratory Data Analysis Output:

The exploratory data analysis (EDA) phase was done to understand the structure, quality, and behavioural patterns present in the Amazon review dataset before applying any modelling techniques. The original dataset consisted of approximately 83,985 reviews. After removing duplicate review IDs and cleaning missing or inconsistent entries, the dataset was reduced to 82,069 valid reviews, ensuring higher data quality for further analysis.

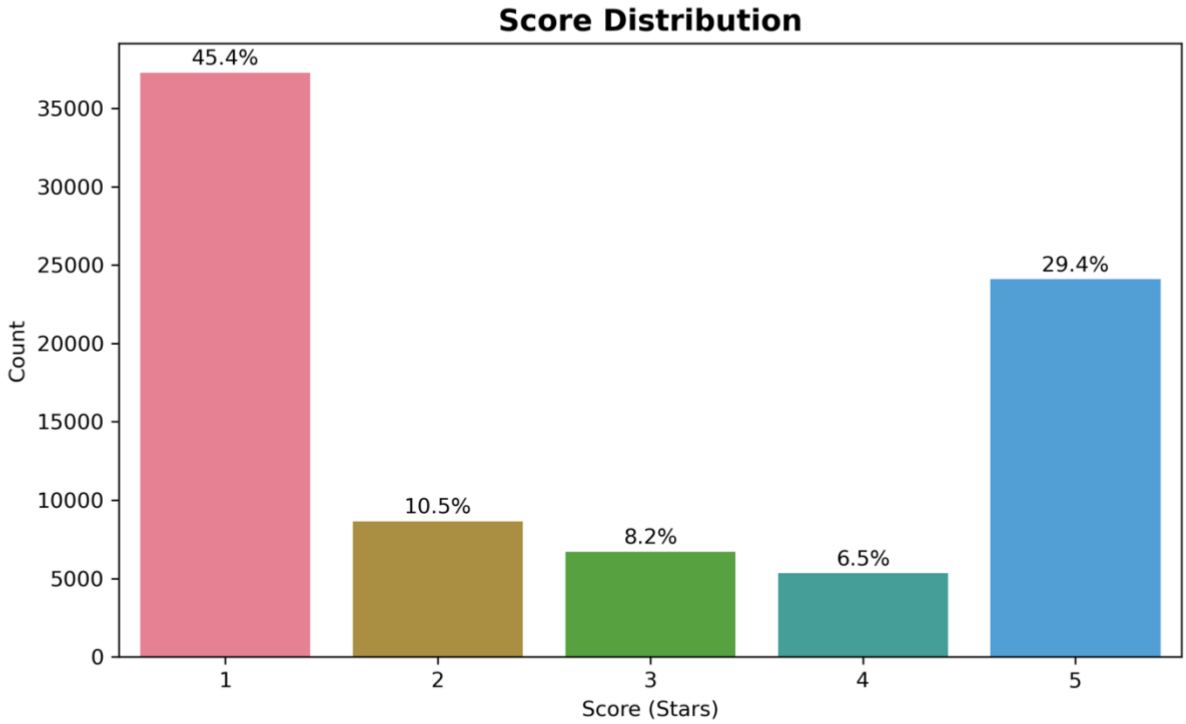

Figure 2: EDA Output - Rating distribution of Amazon app reviews

Several key patterns were identified during this stage. The score-rating distribution showed an uneven spread across star ratings, with a noticeable concentration of higher ratings. For sentiment modelling purposes, 3-star reviews were treated as neutral and excluded, resulting in a sentiment-ready dataset of 75,360 reviews. This dataset contained 45,927 negative reviews (1–2 stars) and 29,433 positive reviews (4–5 stars), indicating a moderate class imbalance.

The analysis of review length and word count revealed that most reviews were relatively short, with only a small proportion of longer, detailed reviews. This pattern suggests that many users provide minimal textual feedback, which can reduce the richness of sentiment information.

To further understand how textual detail relates to user ratings, the relationship between review word count and rating score was analysed.

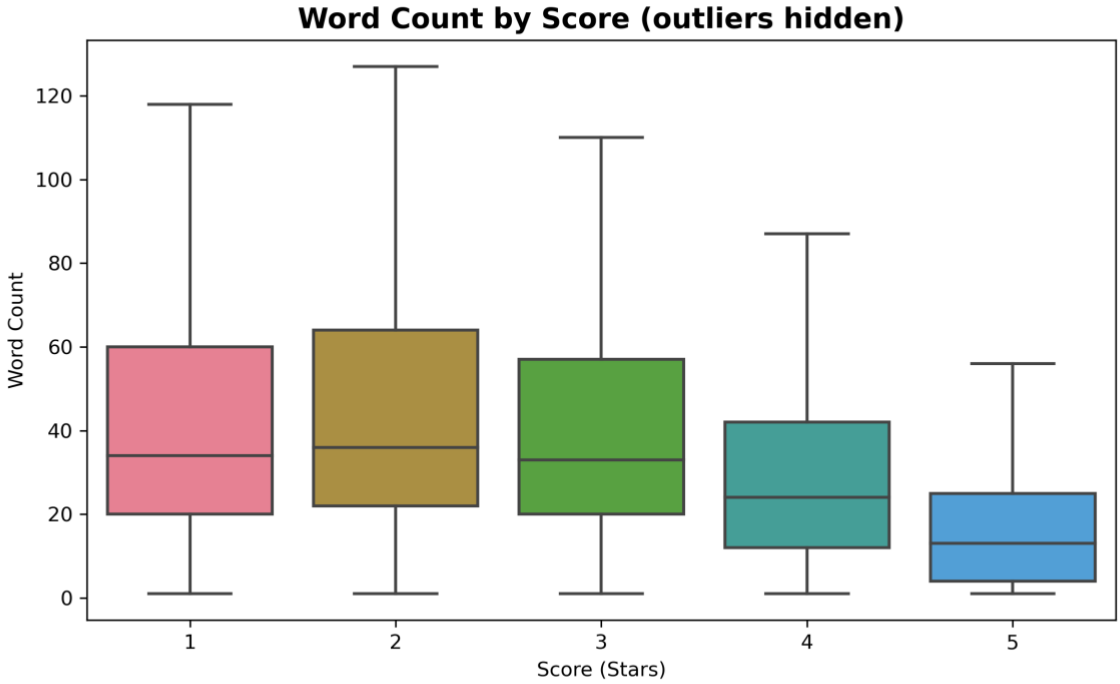

Figure 3: EDA Output - Word count by score

The visualisation shows that shorter reviews are more common across all rating levels, while longer reviews tend to provide more detailed feedback. This suggests that review length alone is not a strong indicator of sentiment polarity, but it remains a useful behavioural feature for anomaly detection.

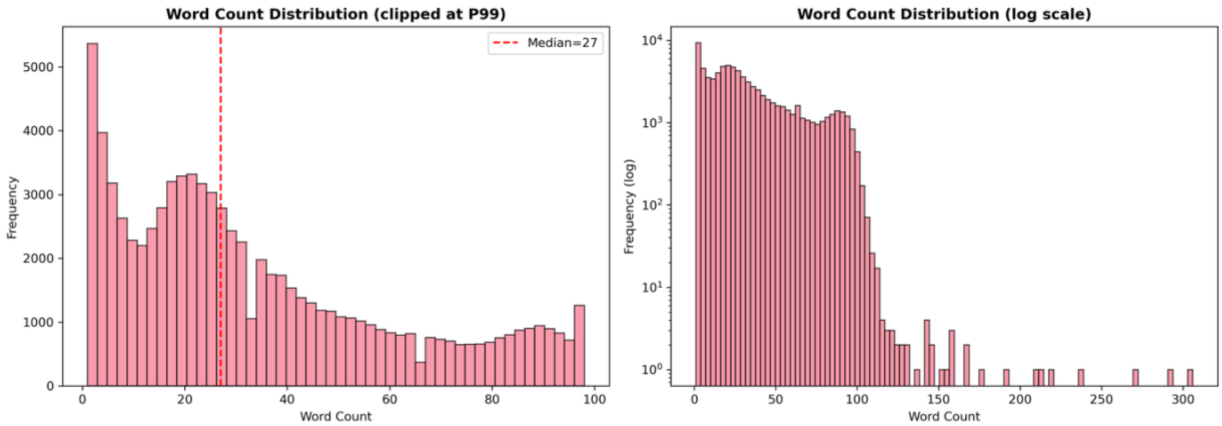

Figure 4: EDA output - Word count distribution

The thumbs-up distribution further supported this observation, with approximately 62% of reviews receiving zero thumbs-up, indicating low user engagement for most reviews.

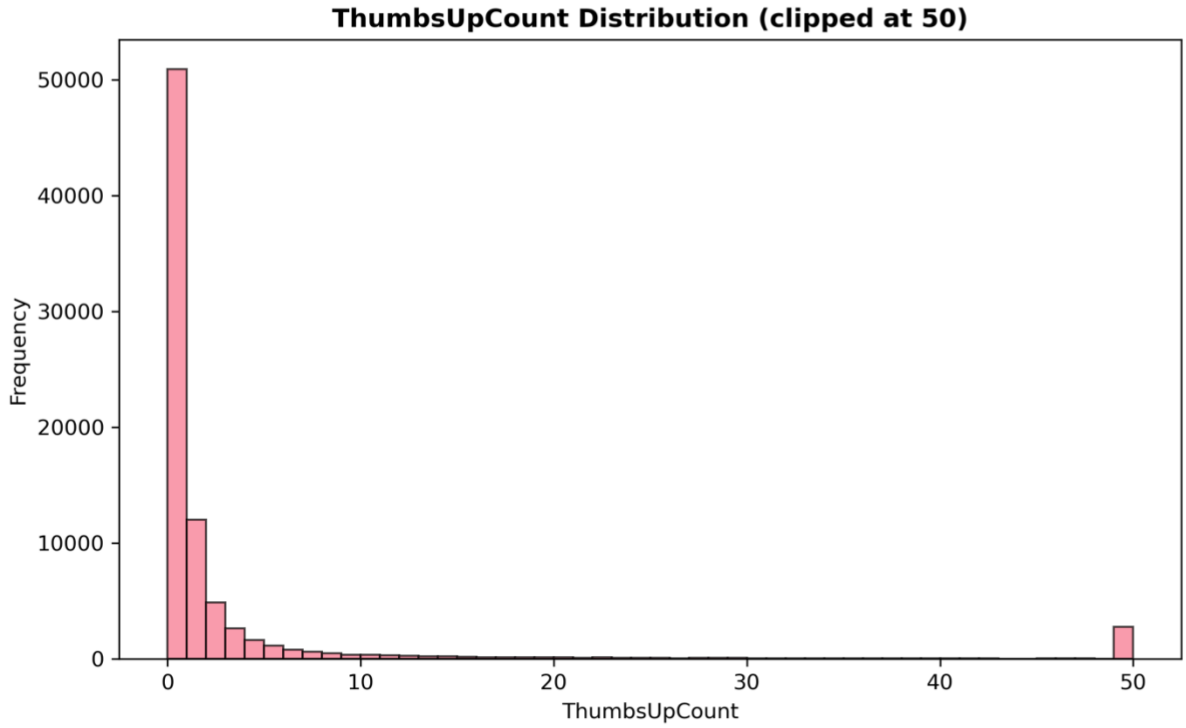

Figure 5: EDA Output - Thumbs-up count distribution

Duplicate text analysis revealed repeated short expressions such as “good”, “great”, and “excellent”, suggesting the presence of low-effort or potentially automated content.

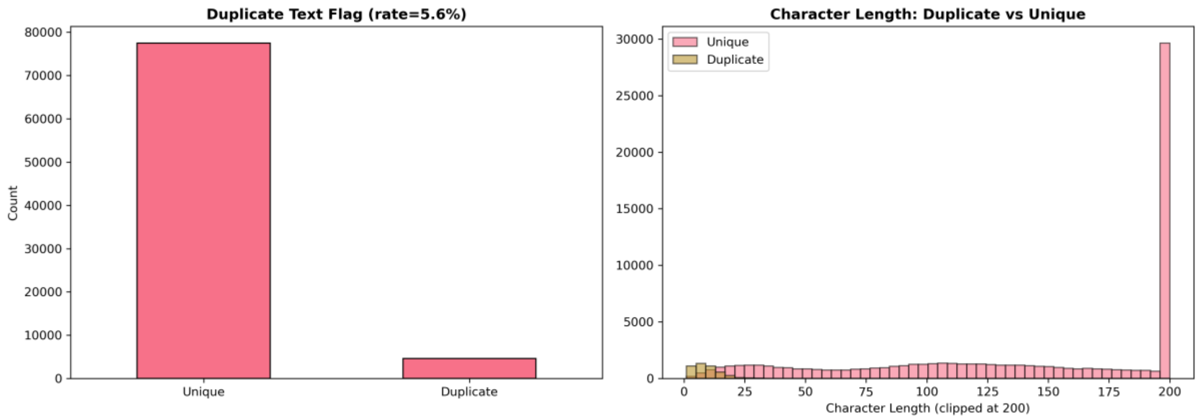

Figure 6: EDA Output - Duplicate text flag and character length

In addition to textual and behavioural features, temporal patterns of review activity were analysed to understand how reviews are distributed over time. The dataset spans from September 2018 to February 2026, with varying levels of review activity across this period. The analysis shows fluctuations in review frequency, including notable peaks where review activity increases significantly on specific dates.

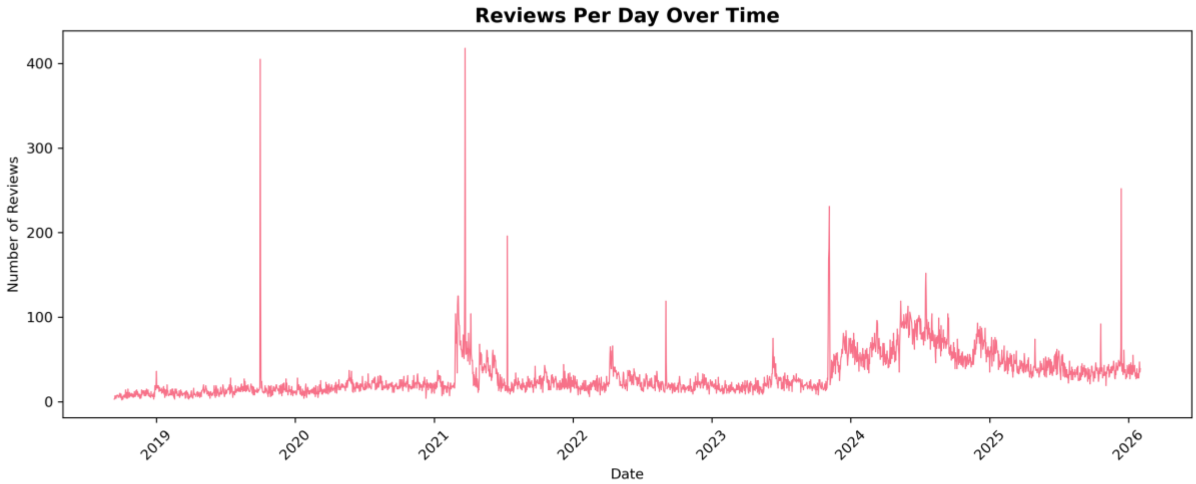

Figure 7: EDA Output - Reviews per day over time

The temporal distribution highlights that review activity is not uniform over time. Certain periods show spikes in the number of reviews submitted, which may indicate events such as product updates, promotional activity, or coordinated review behaviour. These patterns are important for anomaly detection, as unusual spikes in activity can signal potentially suspicious behaviour that requires further investigation.

Additional textual features, including capitalisation ratio, punctuation usage, emoji frequency, and VADER sentiment scores were also examined. These features were later used for anomaly detection and sentiment classification.

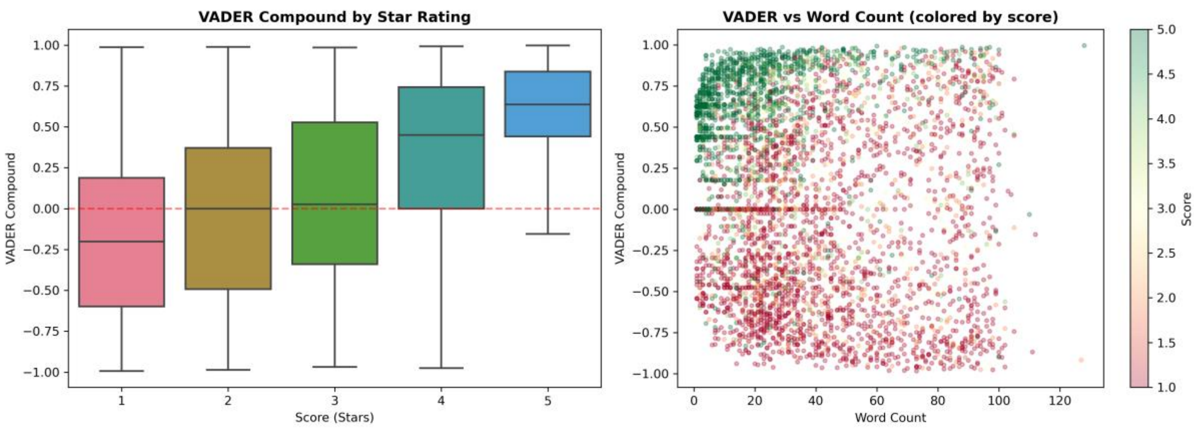

Figure 8: EDA Output - VADER compound by start rating and score

The distribution of reviews across different application versions was also analyses to understand whether user feedback varies with software updates.

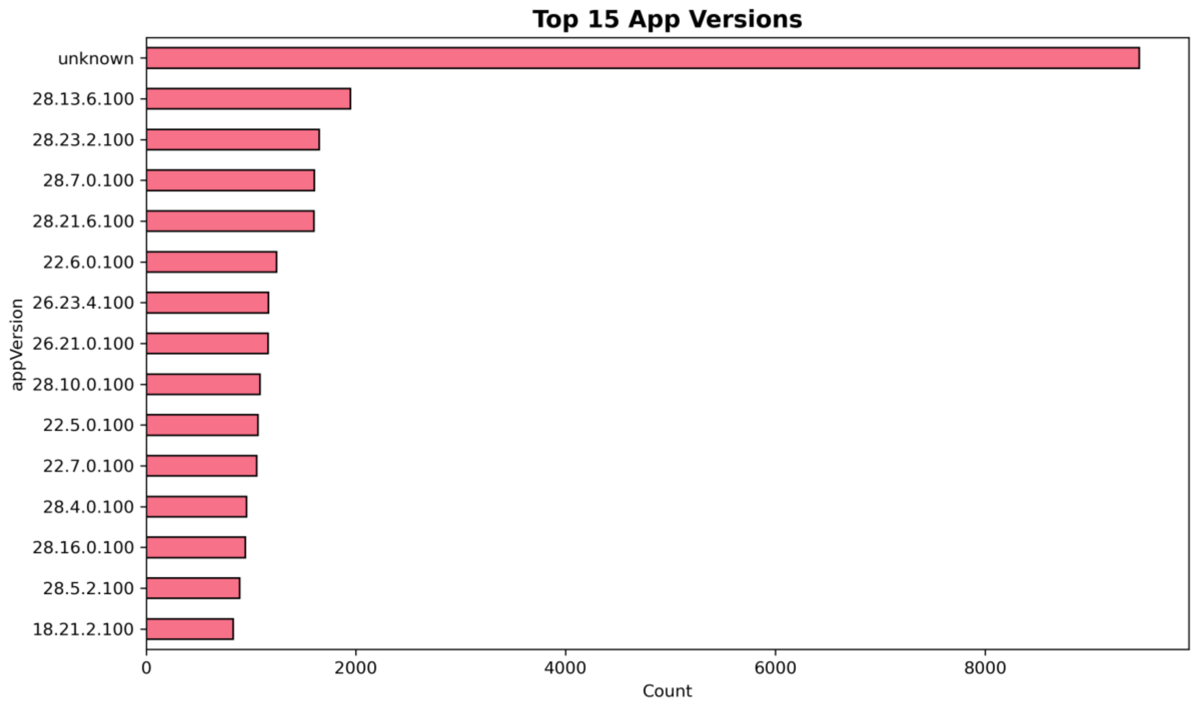

Figure 9: EDA Output - Top 15 app versions

The analysis shows that certain app versions receive significantly more reviews than others. This may reflect periods of increased user activity, updates, or issues affecting specific versions. Such patterns are useful for identifying external factors that influence review behaviour.

Correlation analysis showed strong relationships between certain features, such as character length and word count, as well as moderate relationships between rating scores and sentiment polarity.

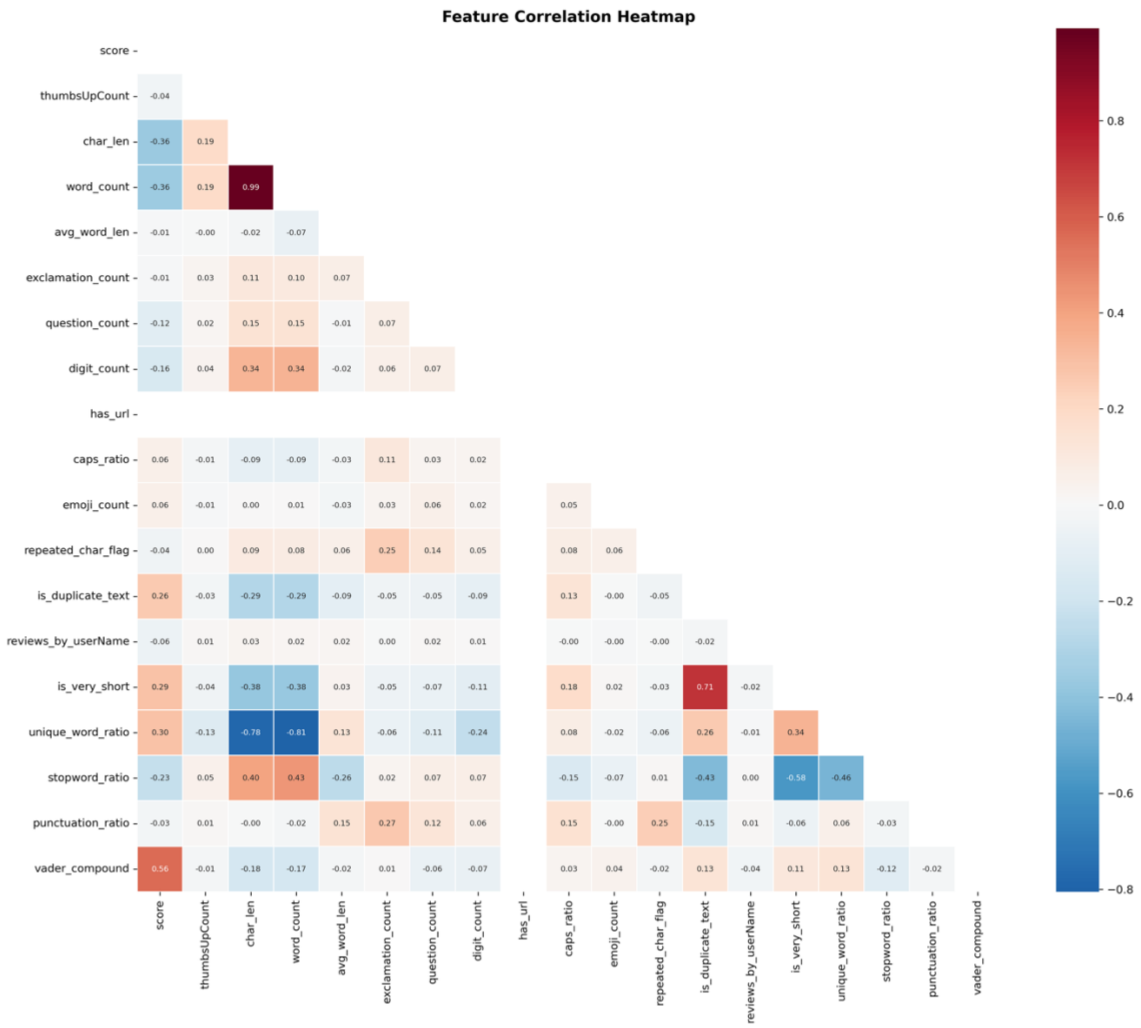

Figure 10: EDA Output - Feature correlation heatmap

These findings confirmed that the selected features captured meaningful behavioural and linguistic patterns within the dataset and provided a strong foundation for subsequent modelling.

Outlier analysis was conducted to identify extreme values in behavioural and textual features that may indicate unusual review patterns.

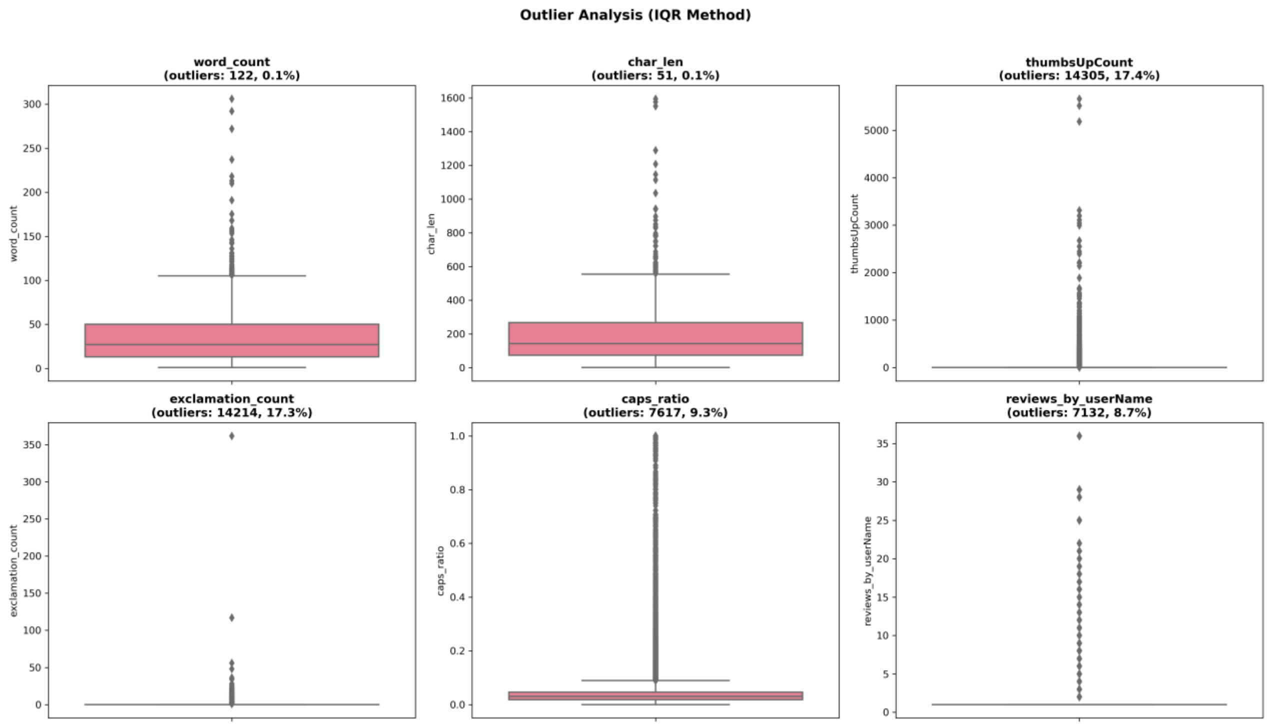

Figure 11: EDA Output (IQR Method)

The presence of outliers in features such as review length, punctuation usage, and rating behaviour suggests that certain reviews deviate significantly from normal patterns. These extreme cases are important inputs for anomaly detection, as they may represent suspicious or non-typical review behaviour.

3. Anomaly Detection (Fake Review Detection):

3.1 Output and Visualisation:

The anomaly detection aimed to identify suspicious or potentially unreliable reviews using both machine learning and rule-based approaches before proceeding with sentiment analysation.

An Isolation Forest model was applied using 21 engineered features, including review length, average word length, rating behaviour, punctuation ratio, duplicate text weight, and sentiment-related metrics. The model identified 8,207 reviews as anomalous, representing approximately 10% of the dataset.

```
Training Isolation Forest on 82,069 samples with 21 features...

Isolation Forest complete:
  Anomalies detected: 8,207 (10.00%)
  Normal reviews:     73,862
  Anomaly score range: [-0.6787, -0.3582]
```

Figure 12: Anomaly Detection Output - IF detection result

This contamination level was selected after testing different thresholds (5%, 10%, and 15%), with 10% providing a balanced detection rate.

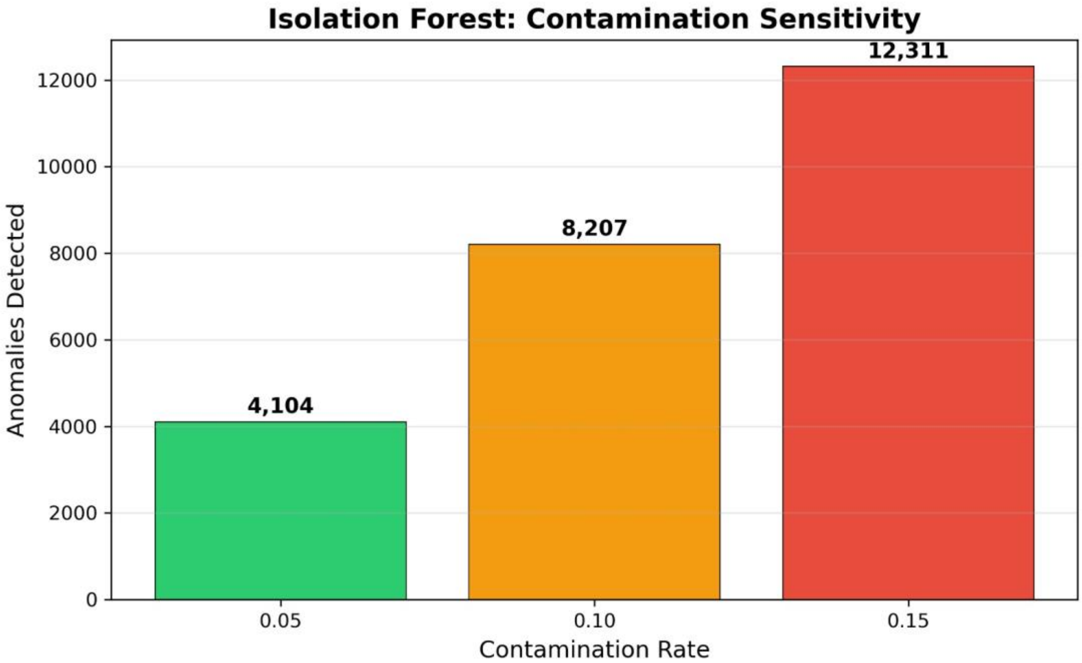

Figure 13: Anomaly Detection Output - Isolation forest contamination sensitivity analysis

In parallel, a rule-based detection system was implemented to capture explicit suspicious patterns. These rules flagged behaviours such as extremely short reviews with extreme ratings, excessive capitalisation, repetitive text, unusual punctuation, emoji overuse, and inconsistencies between rating and sentiment. The rule-based system identified 19,646 reviews that triggered at least one rule.

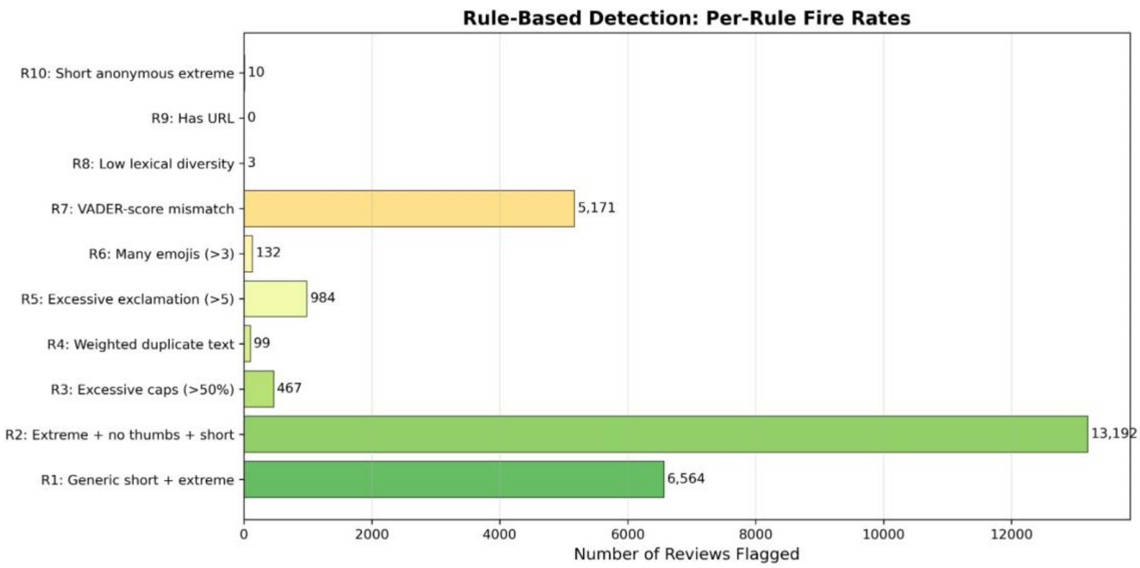

Figure 14: Anomaly Detection Output - Rule-based detection per rule fire rates

To improve reliability, an ensemble voting approach was used. Reviews flagged by three or more rules were considered suspicious, resulting in 2,873 reviews (3.5% of the dataset). This approach reduced false positives and ensured that only consistently suspicious patterns were highlighted.

```
Ensemble Detection Results (11 voters: 10 rules + Isolation Forest):
=================================================================
Conservative (>=4 votes): 50 (0.06%)
Moderate     (>=3 votes): 2,873 (3.50%)
Aggressive   (>=2 votes): 8,218 (10.01%)

Vote distribution:
0    58383
1    15468
2     5345
3     2823
4       48
5        2
Name: vote_count, dtype: int64
```

Figure 15: Anomaly Detection Output - Ensemble voting results for suspicious review detection

Several visualisations were used to support interpretation:

Correlation heatmap showing moderate agreement between rule-based and model-based detection methods.

Anomaly score distribution demonstrating that flagged reviews generally had lower anomaly scores.

PCA visualisation reducing feature space to two dimensions and showing partial separation between normal and suspicious reviews.

Feature importance identifying key contributing variables such as behavioural irregularities and exaggerated writing patterns.

These outputs provided both quantitative and visual evidence of suspicious review behaviour. This also shows that linguistic and behavioural patterns affect more than randomness.

3.2 Evaluation:

The anomaly detection results demonstrate that suspicious review patterns can be identified using a combination of behavioural and linguistic features. The Isolation Forest model successfully captured unusual review patterns, while the rule-based system highlighted explicit suspicious behaviours. However, each method alone had its own limitations. The Isolation Forest flagged a relatively large proportion of reviews, while the rule-based approach risked over-detection due to simple heuristics.

The ensemble voting approach provided a more balanced solution by combining multiple signals, ensuring that only reviews with consistent suspicious characteristics were flagged. This aligns with existing research which suggests that combining multiple detection techniques improves robustness and reduces redundancy dependence on one method alone (Chandola, Banerjee and Kumar, 2009; Aggarwal, 2017).

The PCA visualisation and anomaly score distribution showed that suspicious reviews do not form a separate cluster. Instead, there is partial overlap between normal and anomalous reviews. This is expected, as fake review detection is an unsupervised problem without confirmed ground-truth labels for direct accuracy testing (Liu, Ting and Zhou, 2008; Emmott et al., 2013). As a result, anomaly detection should be interpreted as identifying **suspicious patterns rather than definitive fake reviews.**

Overall, the anomaly detection system provides a meaningful way to assess review reliability. While it cannot guarantee absolute accuracy, it effectively highlights patterns that may distort customer perception and supports more informed interpretation of review data.

4. Sentiment Analysis:

4.1 Output and Visualisation:

The sentiment analysis focused on classifying reviews into positive and negative categories based on textual content. After preprocessing, the sentiment-ready dataset contained 75,360 reviews, with 45,927 negative and 29,433 positive instances.

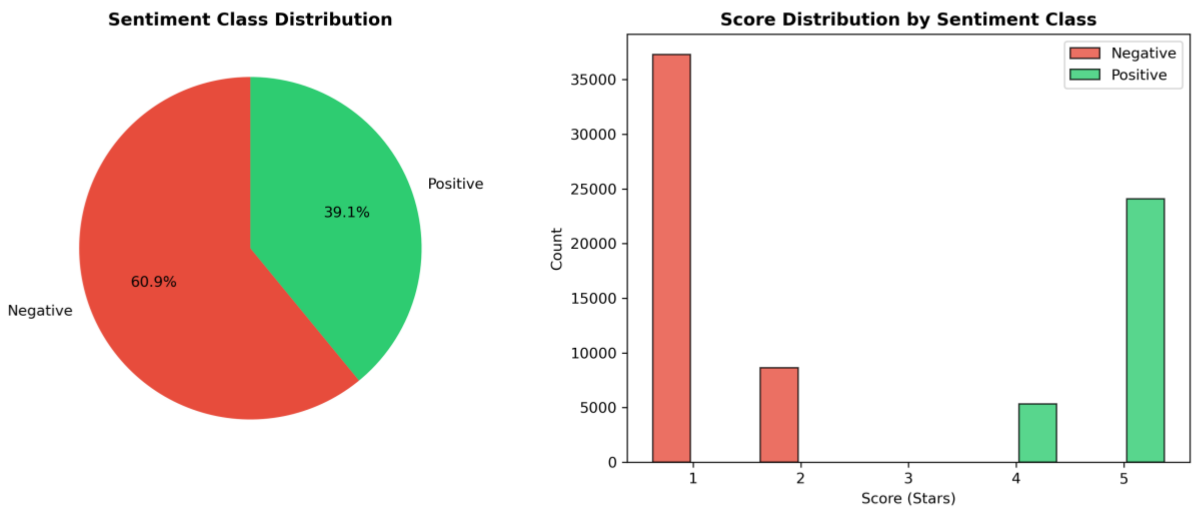

Figure 16: Sentiment Analysis - Output: Sentiment class distribution after neutral review removal

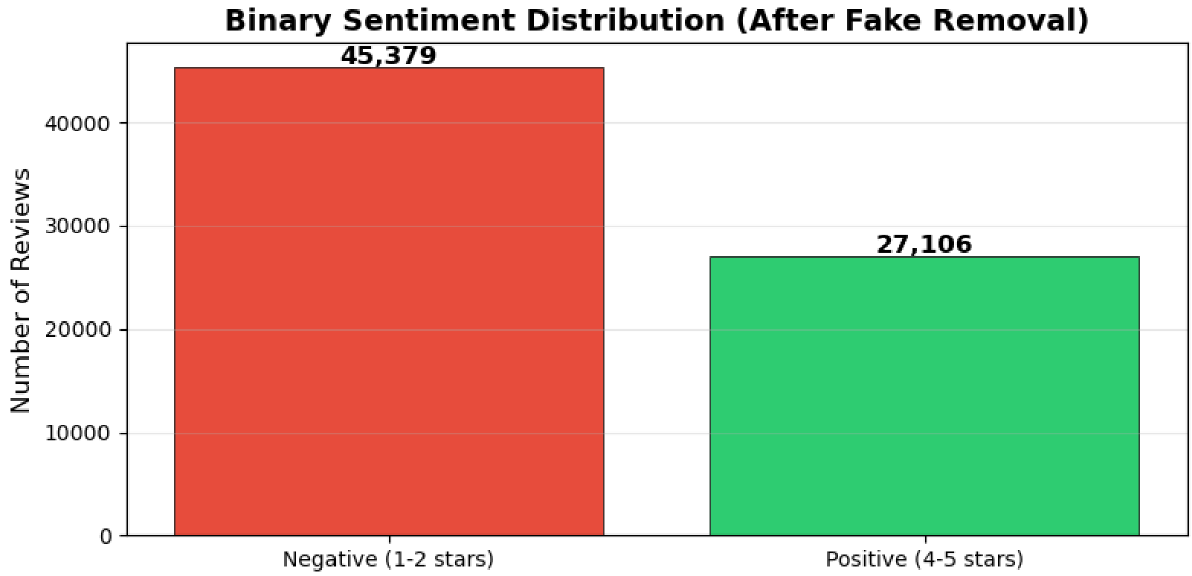

Figure 17: Sentiment Analysis - Output: Binary sentiment distribution after fake removal

Two machine learning models were evaluated: Logistic Regression and Multinomial Naive Bayes. Both models used TF-IDF vectorisation with unigrams and bigrams, limited to 10,000 features.

Figure 18: Sentiment Analysis - Output: Comparing model performance

The results showed that Logistic Regression achieved superior performance, with:

Accuracy: 92.28%

Weighted F1-score: 0.9227

ROC AUC: 0.9678

In comparison, Multinomial Naive Bayes achieved:

Accuracy: 90.49%

Weighted F1-score: 0.9027

ROC AUC: 0.9561

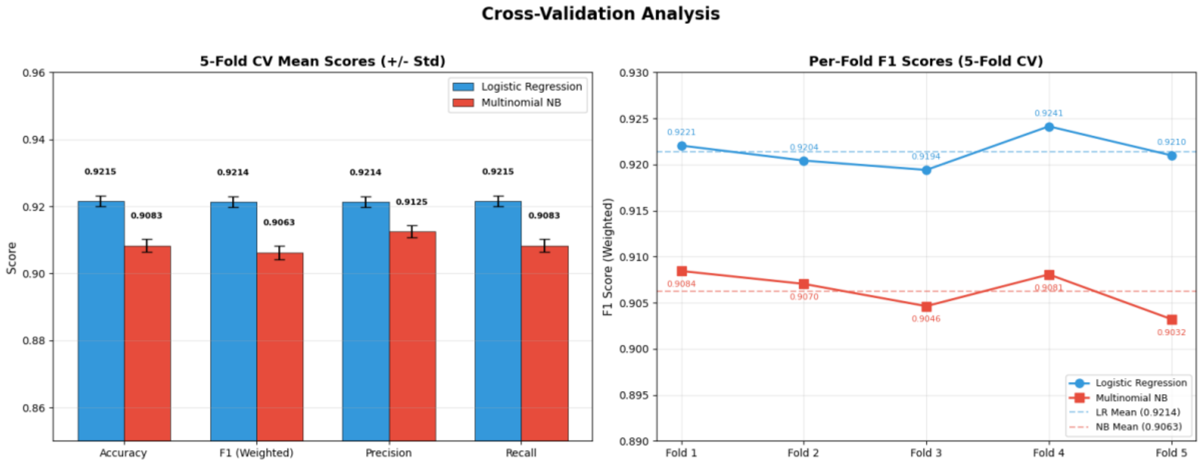

Figure 19: Sentiment Analysis - Output: 5-fold cross valudation analysis

To ensure robustness, stratified 5-fold cross-validation was applied. The Logistic Regression model achieved an average weighted F1-score of 0.9214 ± 0.0016, indicating very low variance across folds and strong generalisation performance.

Several visualisations were used to interpret the results:

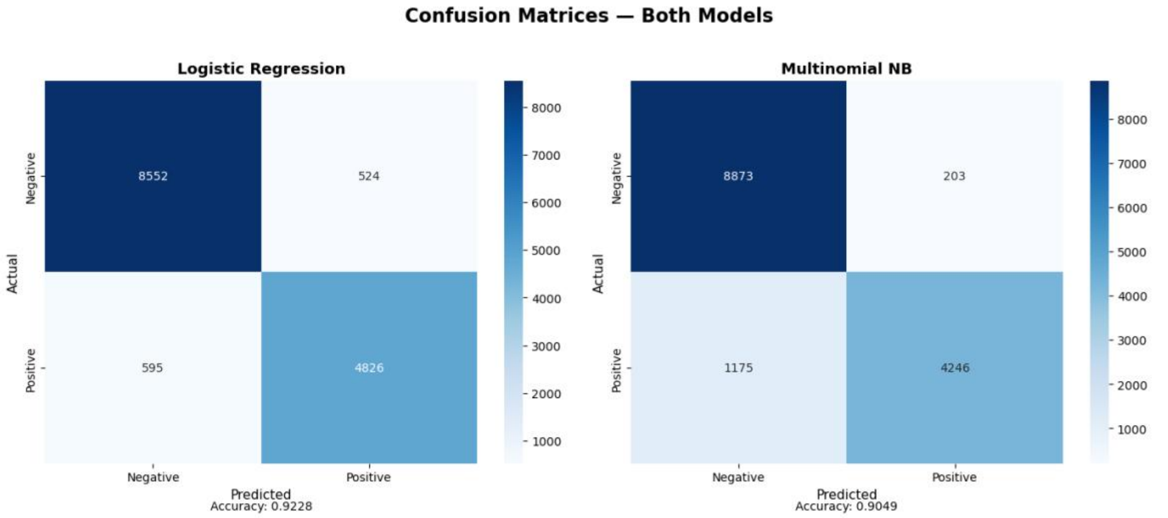

Figure 20matrices for Logistic Regression and Multinomial Naive Bayes: Sentiment Analysis - Output: Confusion

Confusion matrix, showing correct and incorrect classifications

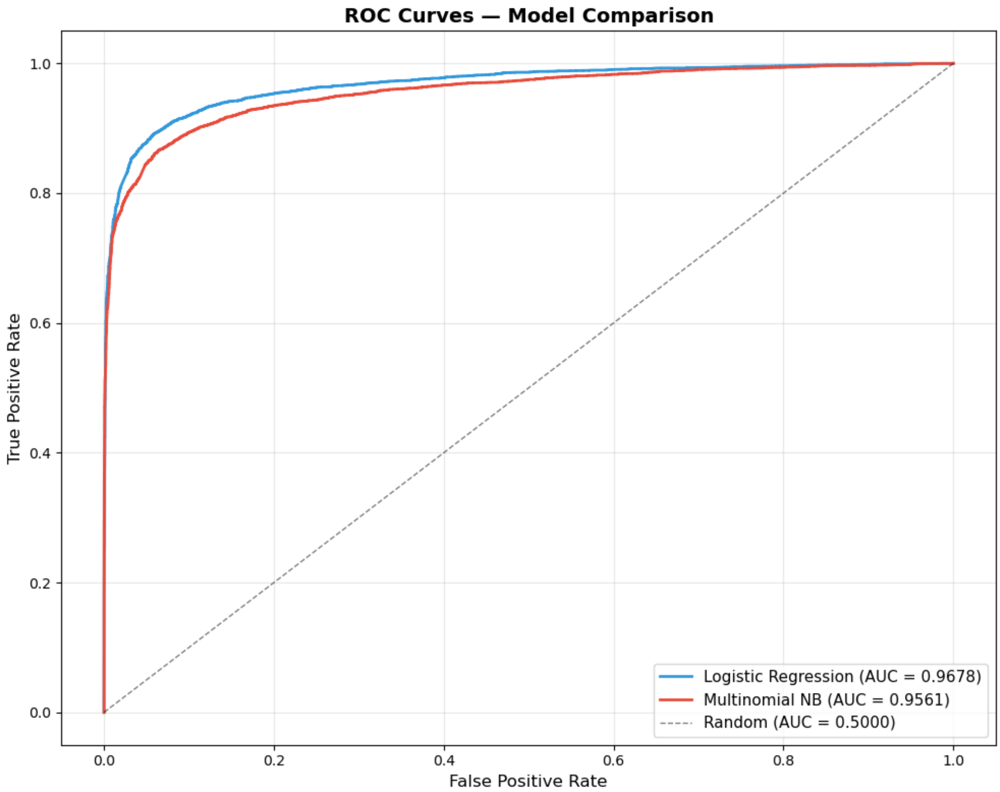

Figure 21: Sentiment Analysis Output: ROC curved for sentiment classification models

ROC curve, demonstrating strong class separability

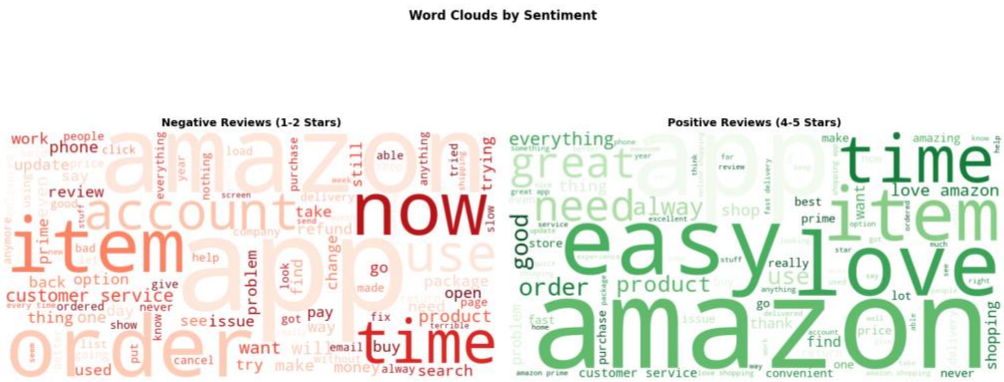

Figure 22: Sentiment Analysis Output: Word cloud for positive and negative reviews

Word clouds, highlighting dominant terms in positive and negative reviews

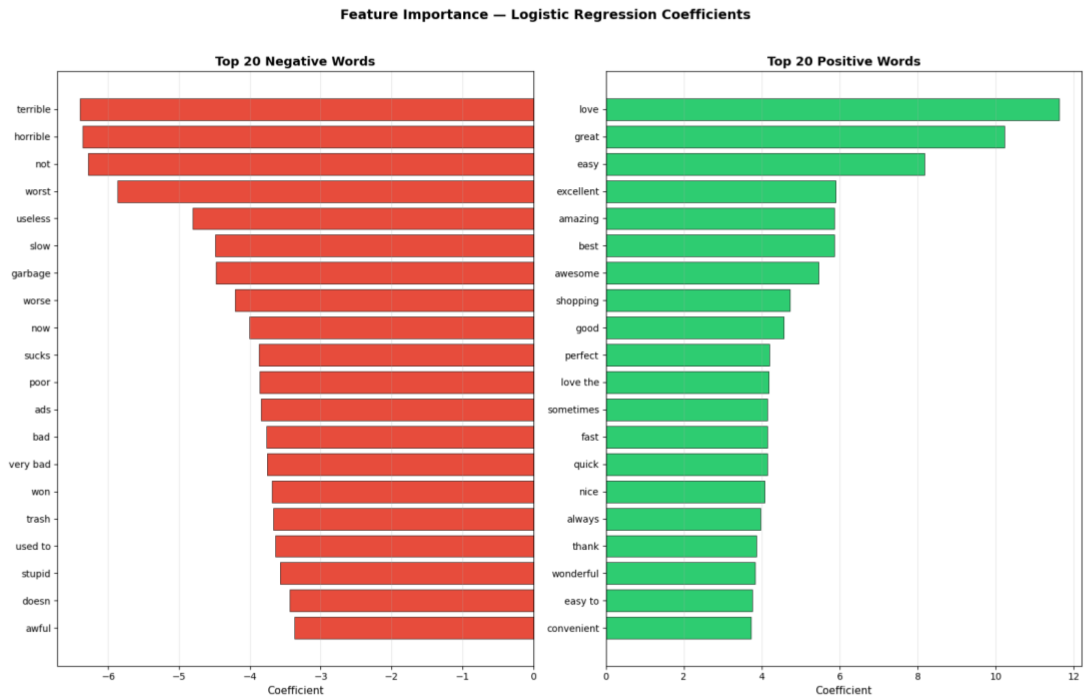

Figure 23: Sentiment Analysis Output: Feature importance on top 20 positive and negative words

Feature importance, identifying key words influencing classification decisions

These outputs confirmed that the model captured meaningful linguistic patterns.

4.2 Evaluation:

The sentiment analysis results indicate that Logistic Regression is the most suitable model for this dataset. It consistently outperformed Multinomial Naive Bayes across all evaluation metrics. The high ROC, AUC value (0.9678) demonstrates strong discrimination between positive and negative classes, indicating that the model is effective at capturing sentiment patterns.

The use of cross-validation further strengthens the reliability of the results, as it confirms that the model's performance is stable and not dependent on a single data split. This is particularly important given the moderate class imbalance in the dataset, as stratified validation helps produce more reliable evaluation results (Kohavi, 1995).

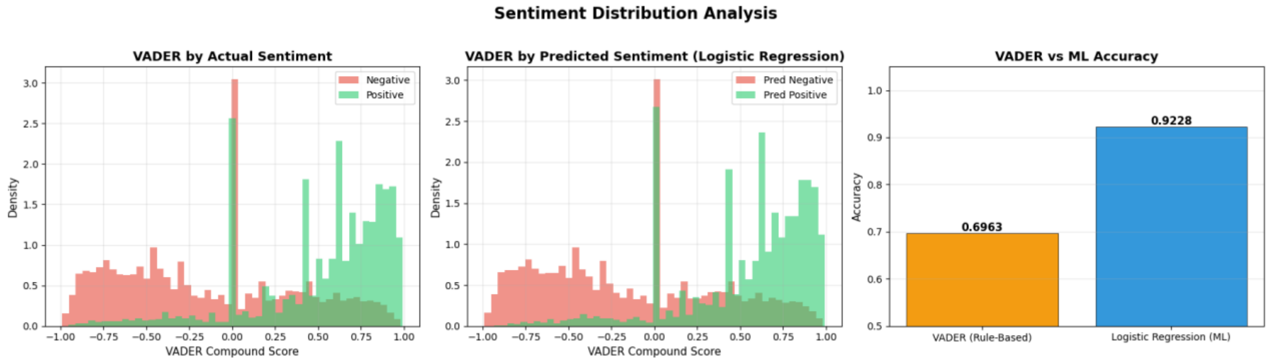

Figure 24: Sentiment distribution analysis

The VADER comparison further shows why the final sentiment classification relied on machine learning rather than rule-based sentiment scoring alone. Although VADER was useful for exploratory analysis and feature engineering, its accuracy was lower than Logistic Regression. This suggests that the TF-IDF-based Logistic Regression model captured review language more effectively and produced stronger sentiment classification results.

The visualisation outputs also support model validity. The word clouds and feature importance analysis show that the model relies on intuitive and interpretable language patterns, such as positive expressions ("love", "great", "excellent") and negative sentiment indicators. This aligns with established sentiment analysis research, which emphasises the importance of checking whether models learn meaningful sentiment-related language patterns rather than relying only on accuracy scores (Pang, Lee and Vaithyanathan, 2002).

Overall, the sentiment analysis component performs effectively and provides reliable classification of review polarity, making it suitable for understanding customer opinions at scale.

5. Integrated System Evaluation:

The combined results from EDA, Anomaly Detection, and Sentiment Analysis demonstrate that the project successfully developed a structured analytical pipeline for improving insight quality in Amazon app reviews.

The EDA stage provided important insights into data quality and user behaviour, highlighting issues such as duplicate reviews, low engagement, and short-text patterns. These findings informed feature engineering and model design.

The anomaly detection system added a layer of reliability assessment by identifying suspicious reviews using both machine learning and rule-based approaches. This is important because previous fake review research shows that deceptive or suspicious reviews often contain behavioural and linguistic signals that can affect the reliability of review-based insights (Jindal and Liu, 2008; Mukherjee et al., 2013). While the absence of ground-truth labels limits definitive validation, the use of multiple evaluation techniques including ensemble voting, PCA visualisation, and score distribution analysis, provides strong supporting evidence.

The sentiment analysis model demonstrated high performance and stability, with Logistic Regression achieving strong classification accuracy and interpretability. The results confirm that meaningful sentiment patterns can be extracted from user-generated review data.

Importantly, the integration of anomaly detection and sentiment analysis supports the overall project aim. By identifying suspicious reviews before interpreting sentiment, the system helps ensure that insights are derived from more reliable data.

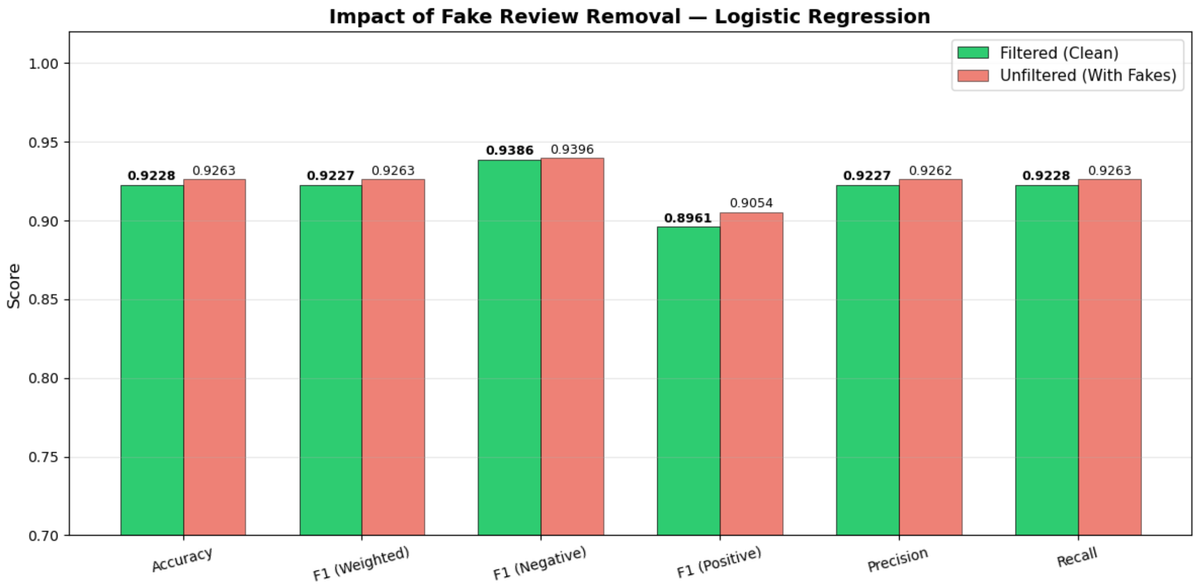

Figure 25: Evaluation: Impact of fake review removal on sentiment classification

However, the results also show that removing suspicious reviews does not necessarily improve classification accuracy, highlighting the distinction between data reliability and model performance**.** The comparison shows that removing suspicious reviews did not significantly improve the Logistic Regression model's classification performance. In fact, the unfiltered data set performed slightly higher across some metrics. This suggests that the anomaly detection is more valuable for improving the reliability and trustworthiness of review interpretation rather than directly increasing sentiment classification accuracy.

In conclusion, the project provides a well-supported analytical framework that combines multiple techniques to enhance the interpretation of online reviews. While limitations remain due to the unsupervised nature of fake review detection, the approach demonstrates both technical effectiveness and critical evaluation, making it suitable as a data artefact for real-world applications.